

FREDERICK REEF, Australia

WMO: 943930

Lat: 20.936S

Lon: 154.401E

Elev: 13

StdP: 101.17

Time Zone: 10.00 (E10)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	19.5	20.0	N/A	N/A	N/A	N/A	N/A	N/A	15.5	20.8	14.5	21.2	8.2	140	0.613

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	1.6	29.1	N/A	28.6	N/A	28.2	N/A	N/A	N/A	N/A	N/A	N/A	N/A	4.2	350

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	(q)
15.7	14.5	13.6	DB	17.9	30.4	0.9	0.6	17.2	30.8	16.6	31.2	16.1	31.5	15.4	31.9	
			WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	24.3	26.7	26.9	26.7	25.3	24.0	22.5	21.8	21.6	22.4	23.3	24.6	26.0
	DBStd	2.12	0.89	0.86	0.80	0.76	0.93	1.01	0.93	0.96	1.06	1.01	0.85	1.02
	HDD10.0	0	0	0	0	0	0	0	0	0	0	0	0	0
	HDD18.3	0	0	0	0	0	0	0	0	0	0	0	0	0
	CDD10.0	5219	518	473	517	460	434	376	364	358	372	414	437	494
	CDD18.3	2177	260	240	258	210	176	126	106	100	122	156	187	236
	CDH23.3	13581	2611	2524	2588	1572	704	118	24	18	109	341	964	2007
	CDH26.7	1201	339	399	308	14	1	0	0	1	0	0	4	136
(14) Wind	WSAvg	7.7	7.2	7.7	8.8	9.3	8.2	8.4	8.2	7.2	6.6	6.9	7.1	6.6
(15) Precipitation	PrecAvg	814	100	139	152	81	56	59	40	38	19	27	32	70
	PrecMax	1223	255	493	425	256	241	172	211	146	58	145	205	242
	PrecMin	400	17	16	8	20	2	11	4	1	1	3	2	4
	PrecStd	231	70	95	122	65	51	44	46	36	16	31	43	54
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	29.6	30.0	29.2	27.2	26.7	25.8	24.6	24.7	25.4	26.4	27.2	29.2
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	2%	DB	28.9	29.1	28.6	27.1	26.2	25.0	23.9	23.8	24.9	25.7	26.6	28.5
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	5%	DB	28.4	28.5	28.2	26.9	25.7	24.4	23.6	23.2	24.5	25.2	26.2	27.9
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	10%	DB	28.0	28.1	27.9	26.6	25.4	24.0	23.2	22.9	24.0	24.9	25.8	27.5
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	2%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	5%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	10%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(35) Mean Daily Temperature Range	5% DB	MDBR	1.6	1.6	1.5	1.5	1.4	1.6	1.4	1.5	1.4	1.4	1.3	1.5
		MCDBR	2.1	2.1	1.8	1.7	1.6	2.1	1.7	1.9	2.0	1.9	1.9	2.2
		MCWBR	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	5% WB	MCDBR	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
		MCWBR	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(40) Clear-Sky Solar Irradiance	taub		0.383	0.385	0.380	0.357	0.341	0.346	0.323	0.330	0.342	0.364	0.371	0.373
	taud		2.592	2.605	2.613	2.648	2.649	2.626	2.680	2.645	2.588	2.546	2.557	2.599
	Ebn at Noon		961	947	928	910	888	863	900	925	950	956	967	971
	Edh at Noon		106	103	98	88	81	80	78	87	99	108	109	105
(44) All-Sky Solar Radiation	RadAvg		7.07	6.55	5.72	5.14	4.34	3.74	4.17	5.13	6.28	7.00	7.40	7.40
	RadStd		0.40	0.64	0.54	0.34	0.26	0.30	0.19	0.29	0.21	0.40	0.53	0.53

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (0 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page